Geophysical Journal (1988) **92,** 223–234

Magnetism of igneous rocks from the Tourmakeady and Glensaul inliers, W. Ireland: mode of emplacement and aspects of the Ordovician field pattern

E. R. Deutsch* and K. M. Storetvedt
Institute of Geophysics, University of Bergen, 5007 Bergen, Norway

Accepted 1987 June 23. Received 1987 June 10; in original form 1987 January 7

SUMMARY

Results of palaeomagnetic and rock magnetic studies are reported mainly for quartz porphyry collected at 18 sites from an Ordovician igneous–sedimentary succession at Tourmakeady and Glensaul in Co. Mayo, Eire. Both magnetite and haematite are shown to be prominent magnetic constituents. The magnetite is usually in the form of large multidomain grains; these are held responsible for a soft remanence component (called the B-magnetization) with D, $I = 000$, $+75$; $\alpha_{95} = 3.7°$, i.e. practically identical to the direction of the present axial geocentric dipole field in the area. Hence, it is inferred that the B-magnetization, which is dominant at most sites, is of recent origin. Haematite, which has a more patchy occurrence, apparently formed through high-temperature oxidation of magnetite. Haematite is the main carrier of the palaeomagnetically significant component, the A-magnetization, which gave well-grouped *in situ* directions (29 samples from 10 sites) with a mean value of D, $I = 182°$, $+6°$, corresponding to a north pole at 31° N, 169° E; dp, dm = 2.1°, 4.2°. Correction for folding along the Glensaul–Tourmakeady arc shows that the A-magnetization is of post-folding age. An earlier suggestion that the rocks represent a Lower Ordovician extrusive suite is discussed and discounted. It is inferred that the quartz porphyries constitute syntectonic intrusions having formed during the Taconic orogeny. These rocks as well as their characteristic magnetization are therefore regarded as late Ordovician in age. It is concluded that a *c.* 50° relative clockwise shift between the mid-Ordovician and late Ordovician–Silurian geomagnetic field axes, previously demonstrated for Scotland, has now been recognized also in Ireland.

Key words: Palaeomagnetism, rock magnetism, Ordovician, W. Ireland

1 INTRODUCTION

Ordovician rocks in the British Isles have provided some useful palaeomagnetic data for estimating the extent of a Proto-Atlantic ocean. Such tests tend to be difficult, however, since they require exact comparisons between synchronously magnetized rocks from opposite sides of a collision suture. Ireland north of the proposed suture (e.g. Phillips *et al.* 1976) is one area where previous Ordovician palaeomagnetic work led to conflicting interpretations. Part of the difficulty arose from the considerable tectonic flexuring in Connemara (Dewey 1963) which accompanied the suggested closure of the Proto-Atlantic. Studies of the Mweelrea ignimbrites of Killary Harbour (Deutsch & Somayajulu 1970; Morris *et al.* 1973) established a consistent mid-Ordovician remanence vector directed southeast, which however was about 30° misaligned in declination with an average vector based on then available results from Ordovician sites in Britain. Briden *et al.* (1973) and Morris (1976) proposed that the discordant direction of the ignimbrites was caused locally by a 30° net horizontal rotation of the South Mayo Trough in the Devonian. However, this 'tectonic' explanation ceased to be viable (Deutsch 1980) after new British and Irish Ordovician results with NW–SE directions (conformable to Mweelrea) became available.

Although block rotations confined to NW Ireland now do not seem to explain the Mweelrea result, it is possible that other palaeomagnetic signatures observed in that region are the result of tectonic distortion. The question is important in the context of Caledonian palaeomagnetism, and in this paper we report results for two Ordovidian inliers exposed at Glensaul and Tourmakeady along a tectonic arc in Connemara (Fig. 1). Between Glensaul and Tourmakeady the regional structure changes from its roughly W–E trend in western Connemara to a SW–NE trend that continues northeastwards roughly along strike with major structures in northern Britain. In a previous palaeomagnetic study of the Glensaul inlier, Morris (1976) obtained a SW-directed remanence and attributed its *c.* 30° misalignment with the characteristic British 'N–S' vector to the bending of an SW–NE structure originally assumed to be linear into its present shape. Morris stated however that AF demagnetiza-

* Permanent address: Geomagnetic Research Laboratory, Department of Earth Sciences, Memorial University of Newfoundland, St John's, Nfld. AIB 3X7, Canada.

Downloaded from https://academic.oup.com/gji/article/92/2/223/752888 by Curtin University Library user on 26 May 2026

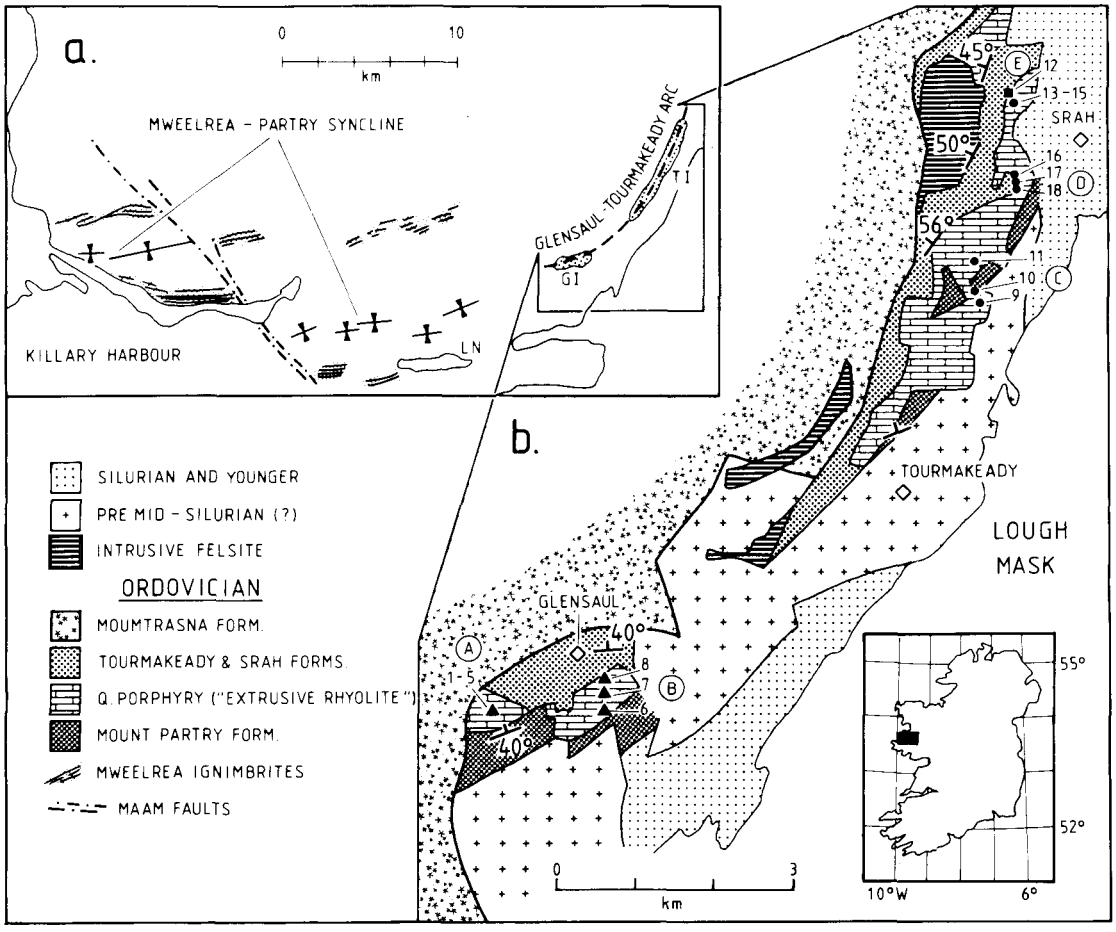


Figure 1. Diagram (a) shows the South Mayo Trough of Connemara, including some major structures, and (b) the Tourmakeady 'volcanic' succession. A–E are palaeomagnetic sampling traverses, each with its strike and dip shown (Table 1). Numbers denote sampling sites in the Glensaul inlier (GI, triangles) and Tourmakeady inlier (TI, quartz porphyry: circles; limestone: square). LN, Lough Nafooey. Geology simplified from Graham *et al.* (1985).

tion, the only laboratory cleaning technique he used, may have been insufficient to isolate an original remanence. Therefore, in the present study we have included thermal demagnetization and we also report for the first time results for the SW–NE-striking Tourmakeady inlier for comparison with W–E-striking Glensaul.

2 GEOLOGY

The so-called Tourmakeady volcanic succession (Graham *et al.* 1985) is exposed along the southeastern margin of the South Mayo Trough (Fig. 1). According to Dewey (1963), this trough developed as an early Ordovician sedimentary basin at the northern rim of a Proto-Atlantic ocean. The Tourmakeady succession comprises the two inliers of Glensaul and Tourmakeady which together form a structure to be called here the Glensaul-Tourmakeady arc (Fig. 1a). This arc lies on the southern limb of the Mweelrea–Partry Syncline, a major asymmetric fold that was created by the Taconic orogeny (Dewey 1961; Ryan & Archer 1978); therefore the folding of the Glensaul–Tourmakeady beds may be assigned an age of mid-Ordovician to early Silurian. However, the age of bending of the Glensaul–Tourmakeady arc itself is unknown.

The Tourmakeady and Glensaul suites were originally described by Gardiner & Reynolds (1909, 1910), who recognized two groups of felsites as the most prominent rocks of this region (Fig. 1b). They named the first group 'intrusive felsites' and the second group 'red' or 'contemporary felsites'. All our igneous samples are from this second group of rocks whose colour varies from pink-purple to grey-green. We shall refer to them as 'quartz porphyry' (Fig. 1b). These rocks occur in outcrops of remarkably variable stratigraphic thickness, ranging from 180 m to 900 m (average *c.* 500 m) in the Tourmakeady sampling area (Ryan 1986). Their mean thickness at Glensaul is about 330 m (Gardiner & Reynolds 1910).

The quartz porphyry is underlain by sedimentary beds of the Mt Partry Formation, some containing Arenig (Lower Ordovician) graptolites (Gardiner & Reynolds 1909, 1910; Dewey *et al.* 1970). Above the quartz porphyry are grit, ash and carbonate beds of the Tourmakeady formation whose age also is cited as Arenig. In both inliers the carbonates occur mainly as (1) dislocated blocks of limestone breccia containing also felsite, and (2) more rarely, bedded limestone. A hard, light-grey, bedded limestone (sampled by us at site 12, Fig. 1b) is faulted against the quartz porphyry exposed nearby (sites 13–15). However, no cases of actual contact between quartz porphyry and any limestone or (with two exceptions: Gardiner & Reynolds

Downloaded from https://academic.oup.com/gji/article/92/2/223/752888 by Curtin University Library user on 26 May 2026

1909) between quartz porphyry and any other adjacent rock have been reported from either inlier.

This lack of contact relations has made it more difficult to settle the important question of the quartz porphyries' mode of formation and age. An extrusive origin would fix their age as Arenig, based on the adjacent fossils. Gardiner & Reynolds (1910) consider the Glensaul felsites (i.e. quartz porphyry in Fig. 1b) to be intrusive, citing petrographic similarities with their 'intrusive felsites' at Tourmakeady. In discussing their 'red felsites' (quartz porphyry) at Tourmakeady, Gardiner & Reynolds (1909) cite four main field observations as relevant: (1) the extremely undulating boundaries of the outcrops; (2) their highly variable thickness (see above); (3) the frequent juxtaposition of felsite and limestone, but absence of overt metamorphism in the latter; and (4) the absence of vesicules and rarity of flow structure in the felsite. They consider points (1) and (3) to favour an extrusive origin of the quartz porphyry and points (2) and (4) an intrusive origin, concluding that on balance the evidence (at Tourmakeady) is pro-extrusive. It seems to us, however, that none of Gardiner & Reynolds' four items of observation provide unequivocal evidence either for or against an extrusive interpretation (except possibly the absence of vesicules, being 'against'). We conclude that, from geological evidence at both Glensaul and Tourmakeady, the mode of formation of the quartz porphyry remains undetermined.

3 SAMPLING AND MEASUREMENT

The rocks were collected in 1983 by one of us (ERD), with guidance in the selection of sampling sites from Dr Paul Ryan of Galway University. A total of 90 hand samples were taken from 18 sites located along five short traverses in the Glensaul and Tourmakeady inliers (Table 1, Fig. 1b). At four of the traverses (A–D) quartz porphyry only was sampled, while the fifth traverse (E) comprises three quartz porphyry sites overlain by a compact bedded limestone of the Tourmakeady formation (site 12). The samples were oriented with a Brunton compass which was checked against a solar compass when possible. Only minor differences between magnetic and solar readings were found and there was therefore no reason to suspect the Brunton readings taken at other outcrops on sunless days. Mean values of strike and dip accurate to 2° or better were obtained at each traverse (Table 1) from adjacent sedimentary beds usually above the quartz porphyry. The dips are confined to the range 40–56°, but there is a progressive decrease in strike along the arc amounting to 50–60° between traverses A and E. As there were no indications of plunging fold axes it was considered safe to reconstruct the original bedding positions by back-tilting about the strike.

Standard cylindrical specimens 2.4 cm in diameter and cut from the 90 samples were used in measurements. A total of 90 specimens (one from each sample) were subjected to thermal demagnetization with a Schonstedt TSD-1 unit operating in residual fields of 5 nT or less. A fresh set of 36 specimens from 36 samples were given alternating-field (AF) treatment with a Schonstedt GSD-1 unit. The intensity of natural remanence (NRM) of the quartz porphyries covers a wide range, $J_n = 10^{-3}$–$5\ A\ m^{-1}$, but in 85 per cent of these specimens J_n is confined between 10^{-2} and $1\ A\ m^{-1}$. Therefore, in most quartz porphyry specimens the remanence was strong enough, even after demagnetization to 1–2 per cent of NRM, to be measured on a Schonstedt SSM-1 spinner magnetometer. The limestones are much weaker ($J_n = 4 \times 10^{-4}$ to $8 \times 10^{-4}\ A\ m^{-1}$) and, in their case and that of the weakest quartz porphyries, the remanence was measured on a CTF cryogenic magnetometer having instrumental noise below $5 \times 10^{-12}\ A\ m^2$.

All the above measurements were made at Memorial University. Additional experimental work, performed at Bergen University, included studies of isothermal remanence (IRM) acquired in fields up to 0.9 T and of thermomagnetic behaviour (J_s–T curves), as well as a microscopic analysis using polished sections.

4 DEMAGNETIZATION RESULTS

The 36 specimens that were subjected to alternating field (AF) demagnetization were treated in steps up to 100 mT (milli-Tesla) peak. Representative examples are depicted in Fig. 2. The AF treatment divided the samples into two groups according to their directional behaviour. The major group is characterized by steep downward inclinations, and the directionally highly stable specimens ($N = 18$), including TY 1–2, 23-A, 26-B, 75-C and 76-A of Fig. 2, define a remanence population with D, $I = 000$, $+75$, $\alpha_{95} = 3.7°$. This magnetization direction is not significantly different from that of the present axial geocentric dipole field (PEF) for W Ireland (Fig. 6). This group of specimens is further described by (a) median destructive fields (MDF) normally below 10 mT, and by the fact that (b) the remaining remanence intensity after 100 mT treatment is generally less

Table 1. Summary of the palaeomagnetic collection.

Traverse*	Inlier	Grid reference†	Sites*‡	Samples‡	Mean strike, dip§
A	Glensaul	266 000/105 000	1–5	TY 1–25	250, 40 N¶
B	Glensaul	265 500/106 000	6–8	TY 26–40	263, 40 N**
C	Tourmakeady	271 000/110 500	9–11	TY 41–55	235, 56 NW**
D	Tourmakeady	272 000/111 000	16–18	TY 76–90	205, 50 W**
E	Tourmakeady	273 000/111 000	12–15	TY 56–75	200, 45 W**

* See Fig. 1.
† From Irish Ordnance Survey, Topographic Map 11 (South Mayo), scale 1:126 270, given to ≈500 m.
‡ Five samples were collected from each site. Samples TY 56–60 (site 12) are limestones; all other samples are quartz porphyry (felsite).
§ Values given are based on 2–9 determinations each.
¶ Determined by the authors.
** Determined by Dr Paul Ryan, Galway University (1986).

Downloaded from https://academic.oup.com/gji/article/92/2/223/752888 by Curtin University Library user on 26 May 2026

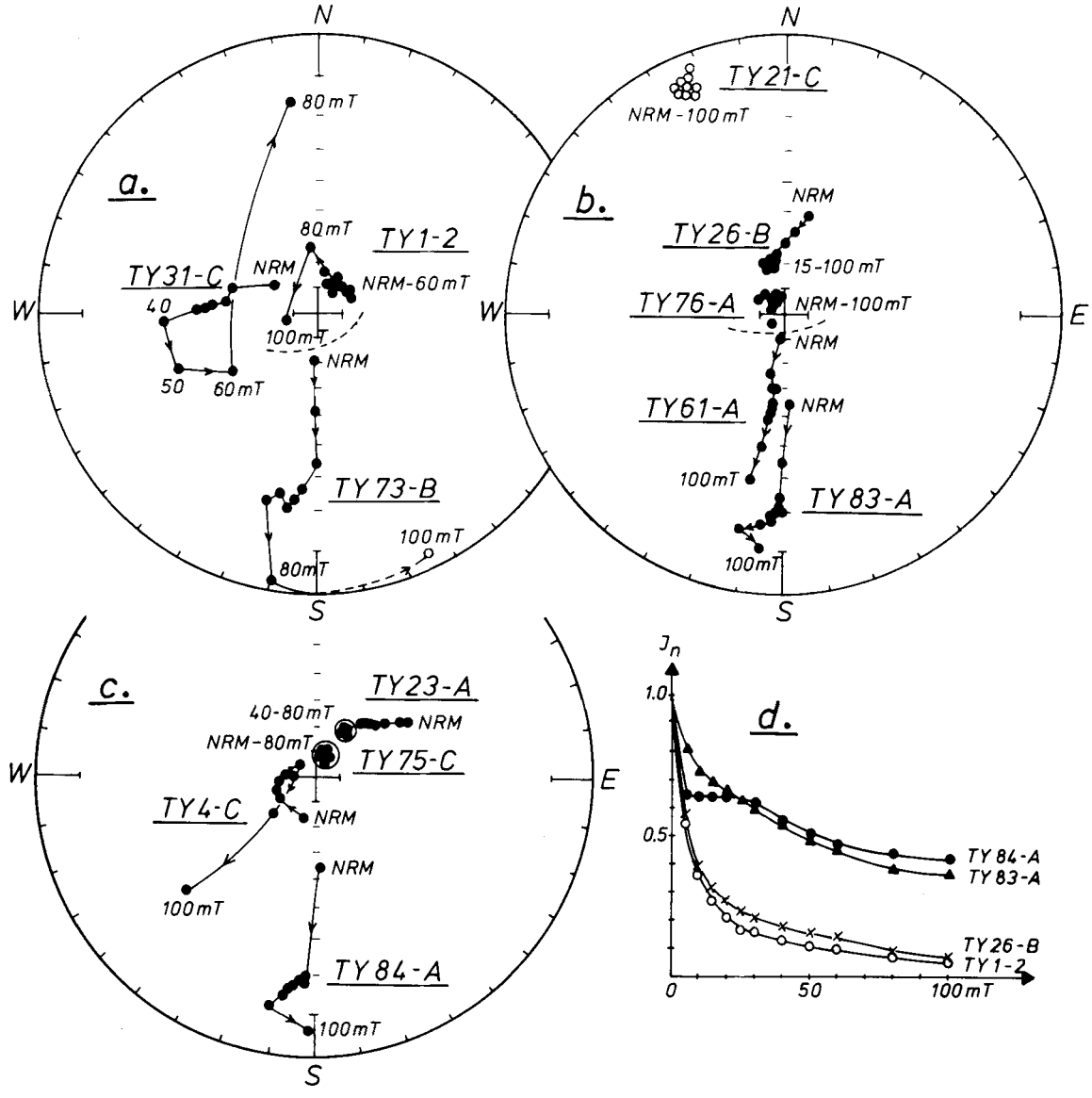


Figure 2. Stepwise AF demagnetization of representative quartz porphyry specimens from the Glensaul inlier (TY 1, 4, 21, 23, 26 and 31) and Tourmakeady inlier (all others). (a–c) Directions projected on a stereographic (Wulff) net. Solid circles denote lower hemisphere; open circles, upper hemisphere. Large circles indicate directions in the field intervals shown and have no statistical significance. (d) Decay of normalized intensity (J_n) for specimens carrying a charactersitic soft B-component (2 lower curves) or hard A-component (2 upper curves).

than 10 per cent of the NRM value (cf. TY 1–2 and 26-B, Fig. 2(d)). These magnetically soft specimens appear to be almost totally governed by a magnetization of recent origin (see also below). We shall refer to this low-coercive force (H_c) remanence as the B-component. This magnetization is the entirely dominant one in rock samples of greenish colour.

The second group of specimens has a characteristic magnetization that is significantly misaligned with the PEF. In the early demagnetization stages the remanence vector moves consistently south, but from say 10 mT onwards, resolution of the components becomes much more difficult and their separation is not accomplished within the 100 mT field employed (e.g. specimens TY 73-B. 61-A, 83-A and 84-A, Fig. 2). This together with directional fluctuations shown by some specimens at higher field steps made it impossible to determine the (reversed) stable end point magnetizations with reasonable precision. However, the overall impression is of the existence of a remanence directed nearly due south and with shallow inclination. One specimen, TY 21-C (Fig. 2b) behaved differently from all others by defining a completely stable direction striking NNW and with shallow upward inclination, apparently representing the normal component of the palaeomagnetic axis in question. This shallow magnetization (of either polarity), which we shall refer to as the A-remanence, was successfully isolated by thermal analysis (see below). The A-remanence is associated with magnetically harder rocks characterized by (a) MDF's ranging between 15 mT and 50 mT and by (b) residual intensities being of the order of 40 per cent of the NRM value after 100 mT treatment (cf. TY 83-A and 84-A, Fig. 2d). The A-magnetization is generally found in rocks of reddish colour.

In terms of component resolution, thermal demagnetiza-

Downloaded from https://academic.oup.com/gji/article/92/2/223/752888 by Curtin University Library user on 26 May 2026

tion proved to be much more powerful than the AF treatment. Again the analysed material (90 specimens) constitutes two entirely different categories: (1) the major group (the greenish rocks) with relatively low blocking temperatures (T_b), which is dominated by a steep, apparently recent *B*-magnetization, and (2) a smaller group (the reddish rocks) with much higher T_bs, for which shallow N–S directed *A*-remanences predominate. Material with unstable magnetic properties (the low H_c, low T_b category) has been found at all our sites except site 12 (limestone). However, the relative proportion of unstable magnetism varies greatly both within and between sites, with the Glensaul collection (sites 1–8) displaying on the whole a much softer magnetization than do the Tourmakeady rocks (sites 9–18) (see Fig. 3). Also, thermal demagnetization of specimens in the soft category (magnetization *B*) show much more unstable directional behaviour, i.e. complex vectorial swings without preferred orientation, than do corresponding AF-treated samples. This pronounced thermal instability is further evidence in favour of a recent viscous origin for the *B*-component. Though practically all specimens analysed carry some steep component of recent origin, the subsequent description of thermal data is confined to samples that have retained a measurable moment of the *A*-magnetization, which apparently is the only component of palaeomagnetic relevance in the present rocks.

Figures 4 and 5 give a broad coverage of thermal demagnetization results. The magnetization properties can frequently be followed to temperatures of 650 °C or higher. This signifies that haematite is an important remanence carrier at least in parts of the Tourmakeady collection. In a small group of samples an apparent univectorial remanence is attained above *c.* 200 °C (e.g. TY 21-A, 60-A, 65-A and 72-B, Fig. 4), but it is commonly found that the

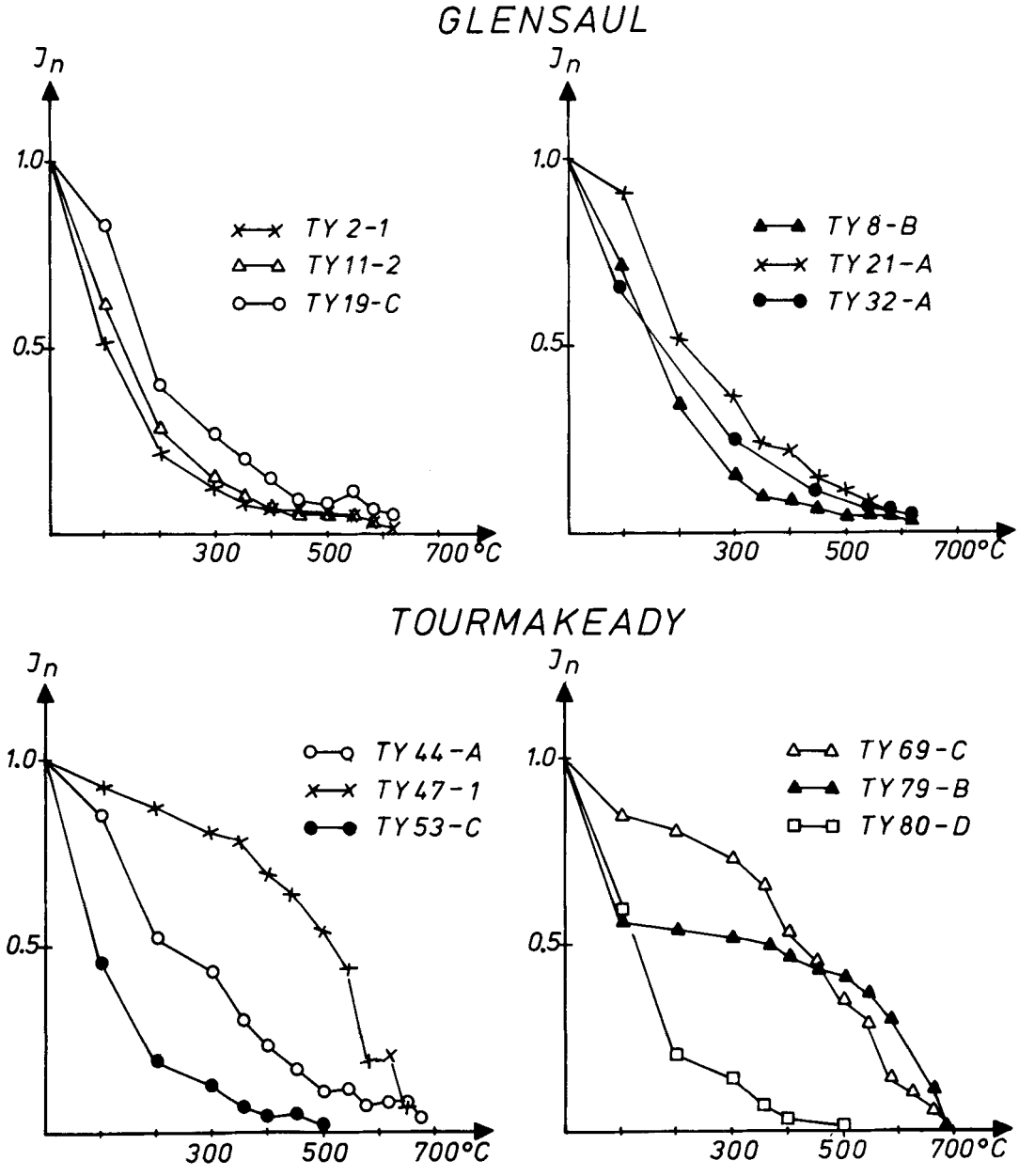


Figure 3. Decay of normalized intensity (J_n) upon stepwise thermal demagnetization of quartz porphyry specimens from Glensaul (sites 1–5 and 7) and Tourmakeady (sites 9–11, 14 and 16).

Downloaded from https://academic.oup.com/gji/article/92/2/223/752888 by Curtin University Library user on 26 May 2026

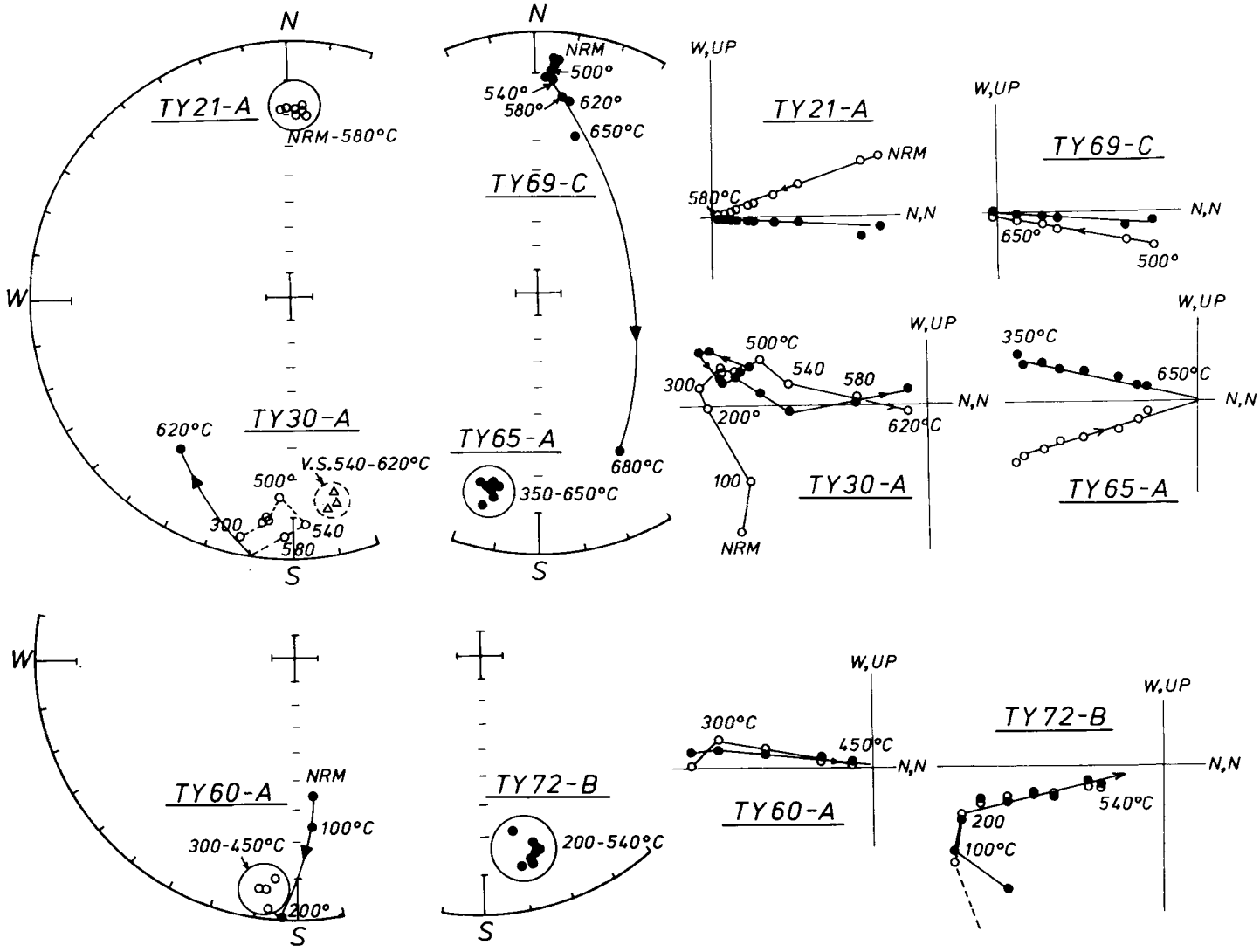


Figure 4. Stepwise thermal demagnetization of six representative specimens from Glensaul (TY 21 and 30) and Tourmakeady (all others). Specimen TY 60-A is limestone, the others are quartz porphyry. (Left side) Stereographic projection of measured directions (circles) and vector-subtracted directions (triangles). Solid symbols denote lower hemisphere; open symbols, upper hemisphere. Large circles indicate directions in the temperature intervals shown and have no statistical significance. (Right side) Corresponding orthogonal vector projections. Open circles are end-point projections of the demagnetization vectors on a N–S vertical plane; solid circles on the horizontal plane.

high-temperature region ($T_c \geq 540$ °C) constitutes superimposed magnetizations. The presence of composite magnetizations is first of all revealed by great-circle paths above 540 or 580 °C (cf. TY 30-A and 69-C of Fig. 4; TY 78-C, 83-C, 84-B, 85-C and 87-B of Fig. 5). None of these examples define stable end-point directions, but the various vector-subtracted directions obtained along the individual remagnetization circles are very well grouped, suggesting non-overlapping stability spectra of the two high-temperature components involved. The mean vector-subtracted directions for individual specimen trends, corresponding to the direction of the erased vector, are also represented by nearly perfect linear high-temperature segments in orthogonal vector diagrams (Figs 4 and 5). The bulk of the *A*-magnetization is of reverse polarity, but a few specimens from sites 5 and 14 are dominated by a normal component. Only from three specimens has a normally directed *A*-magnetization been extracted, but their overall direction is closely antiparallel to that of the reverse group. Nevertheless, the frequent occurrence of remagnetization circles in the high-T_b range of haematite (≥580 °C) suggests that the *A*-magnetization has commonly a dual-polarity build-up. As the most stable components in these cases have not been experimentally isolated, the straight-line segments of orthogonal vector diagrams will, of course, by-pass the origin.

As the AF technique did not prove successful in satisfactorily isolating the *A*-component, only the thermally based results are considered in the further palaeomagnetic assessment. Of the 90 thermally tested specimens, 29 (32 per cent) define the *A*-remanence, while the remaining 61 specimens are dominated by unstable magnetic properties. All 29 *A*-directions correspond to linear segments in orthogonal vector plots, of which several examples are given in Figs 4 and 5. The palaeomagnetic results obtained are of exceptionally high quality, and those Tourmakeady sites

Downloaded from https://academic.oup.com/gji/article/92/2/223/752888 by Curtin University Library user on 26 May 2026

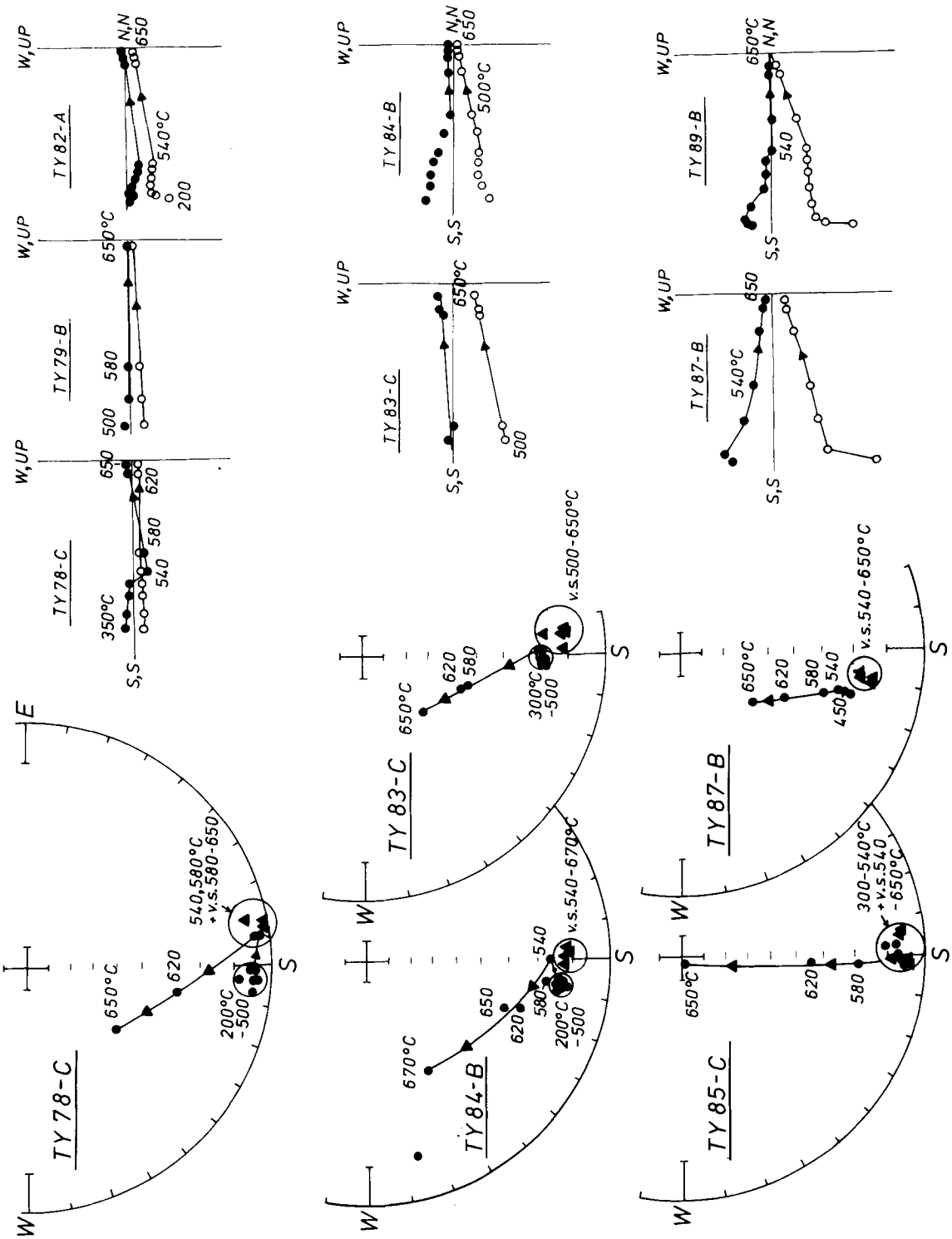


Figure 5. Stepwise thermal demagnetization of quartz porphyry specimens from sites 16–18 (traverse D) in the Tourmakeady inlier. Note that in the majority of cases shown in stereoplots there is a slight difference between the directions of measured quasi-stable magnetization and those of the true remanence (vector-subtracted). The average vector-subtracted direction is represented by linear high-temperature segments in corresponding orthogonal vector plots. Notation as in Fig. 4.

Downloaded from https://academic.oup.com/gji/article/92/2/223/752888 by Curtin University Library user on 26 May 2026

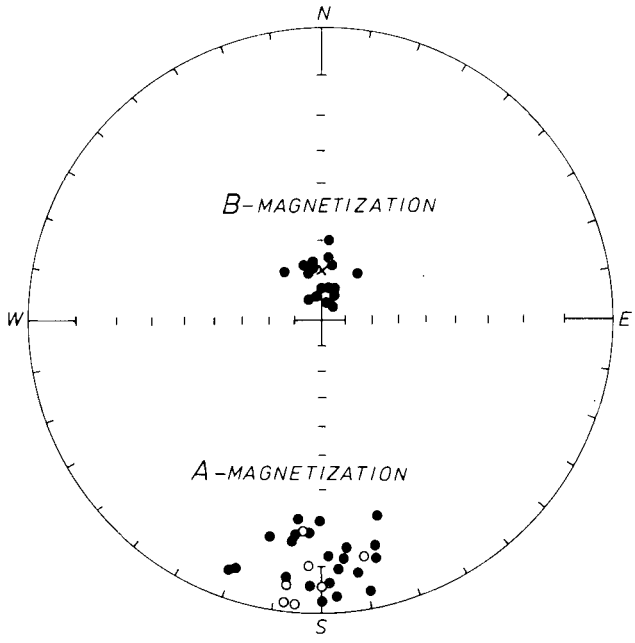


Figure 6. Stereographic plot of individual specimen directions of the two magnetizations. The high-blocking temperature component, magnetization *A*, is calculated from orthogonal vector plots after thermal demagnetization. The 29 specimens concerned (1 specimen per hand sample) are quartz porphyry from Glensaul sites 5 and 6 and Tourmakeady sites 10 and 13–18, and limestone (site 12) from Tourmakeady. The polarity is reverse, except for three normal specimens (2 Glensaul, 1 Tourmakeady) whose directions have been inverted. For comparison with the soft *B*-magnetization the direction of the axial geocentric dipole field is shown by a cross. For overall statistics, see Tables 2 and 3.

Table 2. Overall *in situ* magnetization data for the Glensaul–Tourmakeady arc. *D*, *I* are the mean declineation and inclination for remanence group specified. The mean direction of the *A*-magnetization is calculated from *N* specimen unit vectors (1 specimen per sample) after thermal demagnetization (see text). Of the 29 samples, 24 are quartz porphyry and five are limestone. The directions of three specimens with normal polarity were inverted for calculating *D* and *I*. The parameters of the *B*-magnetization are based on directionally highly stable specimens ($N = 18$) after AF treatment. k = Fischer's precision parameter; α_{95} = radius of the 95% circle of confidence; dp, dm = semi-axes of the 95% confidence oval.

Group	*N*	*D*	*I*	*k*	α_{95}	Pole	dp, dm
A	29	182	+6	38	4.2	31 N, 169 E	2.1, 4.2
B	18	000	+75	80	3.7	82 N, 351 E	6.0, 7.1

which have a preponderance of reddish-coloured rocks have provided the majority of those results.

Out of a total of five specimens per site subjected to thermal demagnetization, the number of specimens for which *A*-magnetizations were retrieved is as follows: site 5:2; site 6:3; site 10:3; site 12:5; site 13:3; site 14:1; site 15:2; site 16:2; site 17:4 and site 18:4 specimens. From this, one sees that the Glensaul sites (nos. 1–8) provide much less promising palaeomagnetic material than do the Tourmakeady sites. The limestone (site 12) is an exception in this study in that, despite its moderate blocking temperatures (e.g. TY 60-A, Fig. 4), the *A*-magnetization was readily obtained in all specimens analysed. The two magnetization components (*A* and *B*) are clearly distinguished in terms of magnetic mineralogy as will be shown later. The *B*-component has apparently no palaeomagnetic interest and will not be discussed further in that context.

Figure 6 shows the distribution of *in situ* remanence directions for the two magnetizations. Overall palaeomagnetic results are listed in Table 2. The markedly different strike directions for Glensaul and Tourmakeady should provide a basis for an effective fold test, but unfortunately there are only five results from Glensaul. However, despite the uneven distribution of results, the stepwise structural unfolding leads to progressive increase in dispersion (cf. Table 3). Thus the fold test is negative, and the results suggest therefore that the *in situ* directions are those of relevance for further palaeomagnetic considerations.

5 MAGNETIC MINERALOGY

Microscopic examination of polished sections unveiled a pronounced difference between rocks of the two inliers. Analysed sections from Glensaul (sites 4, 5, 7) are characterized by large grains identified as magnetite. These particles may be up to 700 μm across, with average grain sizes in the range 100–300 μm. However, these particles exhibit severe fracturing. This penetrative deformation has sometimes reduced the effective magnetic grain size down to the order of 20 μm. While the bulk of the magnetite content is in the original large grains, the matrix also contains some magnetite, but on a scale of a few micrometres. In the interior of large magnetite particles a few of the subdivided regions constitute haematite. These haematite 'grains', which define very sharp outer boundaries, have clearly evolved after the mechanical fragmentation.

Table 3. Site mean analysis. The structural correction (b) is based on data from Table 1. Statistical parameters as for Table 2.

(a) *In situ* site mean parameters

Site	*N*	*D*	*I*	*k*	α_{95}
Glensaul					
5	2	178	15		
6	3	174	−1	30	15
Tourmakeady					
10	3	196	11	48	12
12	5	186	−5	71	7
13	3	177	17	30	15
14	1	183	−10		
15	2	175	9		
16	2	175	3		
17	4	176	10	351	4
18	4	183	14	79	8

(b) Overall site analysis and fold test

Group	*N*	*D*	*I*	*k*	α_{95}
in situ	10	180	6	51	6.2
25% unfold.	10	182	12	44	6.7
50% unfold.	10	184	18	35	7.5
75% unfold.	10	187	23	27	8.5
100% unfold.	10	192	28	20	9.8

Downloaded from https://academic.oup.com/gji/article/92/2/223/752888 by Curtin University Library user on 26 May 2026

Polished sections from Tourmakeady specimens (sites 10, 13, 18) show the same overall pattern with respect to grain sizes and fracturing as does Glensaul, but magnetite has much more extensively been replaced by haematite and secondary silicates. Remains of magnetite are frequently identified only in the interior of the originally large magnetite particles. The matrix reveals a variable but generally very small amount of opaques. The observations suggest that haematite post-dates the mechanical fragmentation of the grains.

From the above evidence we associate greenish rocks with a relatively low state of oxidation of the oxide mineralogy, while the reddish rocks have a high haematite content apparently formed through oxidation of magnetite. We must stress, however, that the haematite content observed in the Tourmakeady rocks is highly variable even over volumes much smaller than a sampling site.

The sharp grain boundaries of haematite and the apparent lack of low-temperature alteration features, i.e. development of haematite as irregular zones along the outer rims and along cracks of individual magnetite particles, suggest that oxidation took place at high temperatures. The extensive fracturing of the larger grains is a general phenomenon also when haematite is absent, and maghaemite has not been identified. Thus, the fragmentation of the magnetite grains is unlikely to be due to volume changes associated with oxidation. As the large magnetite particles must have existed prior to magma emplacement, we believe that the mechanical disruption of these grains was caused by tectonic stress fields acting during emplacement or at the initial cooling stage.

Stepwise thermal demagnetization curves (Fig. 3) demonstrate very well the variability in oxide content. The Glensaul samples are dominated by a low-T_b magnetization and haematite can be confirmed only sporadically as a remanence carrier. The Tourmakeady material shows much greater variability in thermal stability, and within the same location the blocking temperatures may vary from low to very high values (e.g. TY 79-b and 80-D, site 16, Fig. 3). This thermomagnetic variation is clearly correlated with the colour of the samples. In the Tourmakeady collection, in which reddish rocks predominate, the presence of haematite as a remanence carrier is very well exposed.

As a further test for magnetic minerals, we measured isothermal remanence (IRM) and saturation magnetization versus temperature (J_s–T) of rocks from both inliers. Results from six samples have been combined in Fig. 7. For samples TY 18 and 21 (Glensaul sites 4 and 5), both the IRM saturation field (*c.* 300 mT) and coercivity of remanence are relatively low ($H_{cr} < 40$ mT, Fig. 7a, b). For TY 18, the J_s–T heating curve (Fig. 7f) shows a single Curie point at around 580 °C. These results indicate that magnetite is the only significant magnetic mineral in these Glensaul samples.

The Tourmakeady results are more varied. Samples 78 and 89 (sites 16 and 18, Fig. 7i, h) have Curie points near 680 °C. Furthermore, TY 89 has large H_{cr} and, in a 900 mT field, shows no approach to saturation (Fig. 7e). Therefore, in both samples, haematite seems to be the main magnetic material. This is consistent with the evidence from optical observation and demagnetization studies (e.g. Fig. 5). For sample 48 (site 10), a striking two-component trend is shown both by the IRM curves (Fig. 7c) and the J_s–T curves (dual Curie points in Fig. 7g), indicating the presence of both magnetite and haematite. A similar dual magnetite/haematite trend appears in the thermal decay curve of specimen TY 47-1 (Fig. 3) which is also from site 10. Therefore, in the case of both the Tourmakeady and Glensaul quartz porphyry, the IRM and J_s–T results are in all respects concordant with the results of the microscopy and demagnetization studies.

The limestone (site 12) was too weak for measuring T_c from J_s–T curves, but sample 57 (Fig. 7d) yielded two-component IRM curves; these are similar to 'intermediate'-type IRM curves reported for limestones (Lowrie & Heller 1982) which suggest that magnetite and haematite occur together. In the present case this means that probably only the magnetite fraction contributed significantly to the NRM, since all our limestone samples have effectively lost their remanence by *c.* 450 °C.

6 INTERPRETATION

The foregoing evidence suggests that the original magnetic mineralogy of the quartz porphyry basically consisted of large grains of pure magnetite. Such multidomain particles are extremely susceptible to viscous processes and hardly able to retain any early-stage cooling remanence. Mechanical fragmentation of the larger grains, probably brought about by tectonic stress-fields, apparently did not produce grain sizes small enough to provide any significant improvement in magnetic stability. We infer that these large magnetite particles are the carrier of the soft *B*-component. However, a variable amount of haematite seems to have formed through oxidation of original magnetite at high temperature. The *A*-component was usually the result of this oxidation process. The ancient remanence directions acquired through haematite magnetization show predominantly reverse polarity, yet the dual-polarity structure commonly observed (Figs 4 and 5) is indicative of a relatively slow haematization/magnetization process. This suggests that the time of magnetization acquisition was dependent on microenvironmental oxidizing conditions, which in turn should imply that for computation of overall palaeomagnetic parameters individual specimen results are more relevant than site mean directions.

As we noted earlier it is the *in situ* remanence directions that should be considered as palaeomagnetically relevant. This would imply that the *A*-magnetization was acquired after the onset of the main phase of local deformation (Taconic) and so must be mid-Ordovician or later in age. We shall now discuss more closely the question of magnetization age of the rocks in a palaeomagnetic context, and then deal with their possible mode of emplacement.

The *in situ* mean direction for the *A*-component (182°, +6°), when inverted, corresponds to a north pole at 31° N, 169° E; dp, dm = 2.1°, 4.2° (Table 2). In Fig. 8(a), this pole (TY) is plotted along with five published poles (Table 4) of cited mid-Ordovician to Silurian age from areas north of the presumed Caledonide suture. An outstanding feature of our model is the *c.* 50° longitudinal separation between the Middle–Upper Ordovician and Silurian poles, each subset corresponding to a characteristic palaeo-declination axis (Fig. 8b, see caption for details). This two-axis scheme

Downloaded from https://academic.oup.com/gji/article/92/2/223/752888 by Curtin University Library user on 26 May 2026

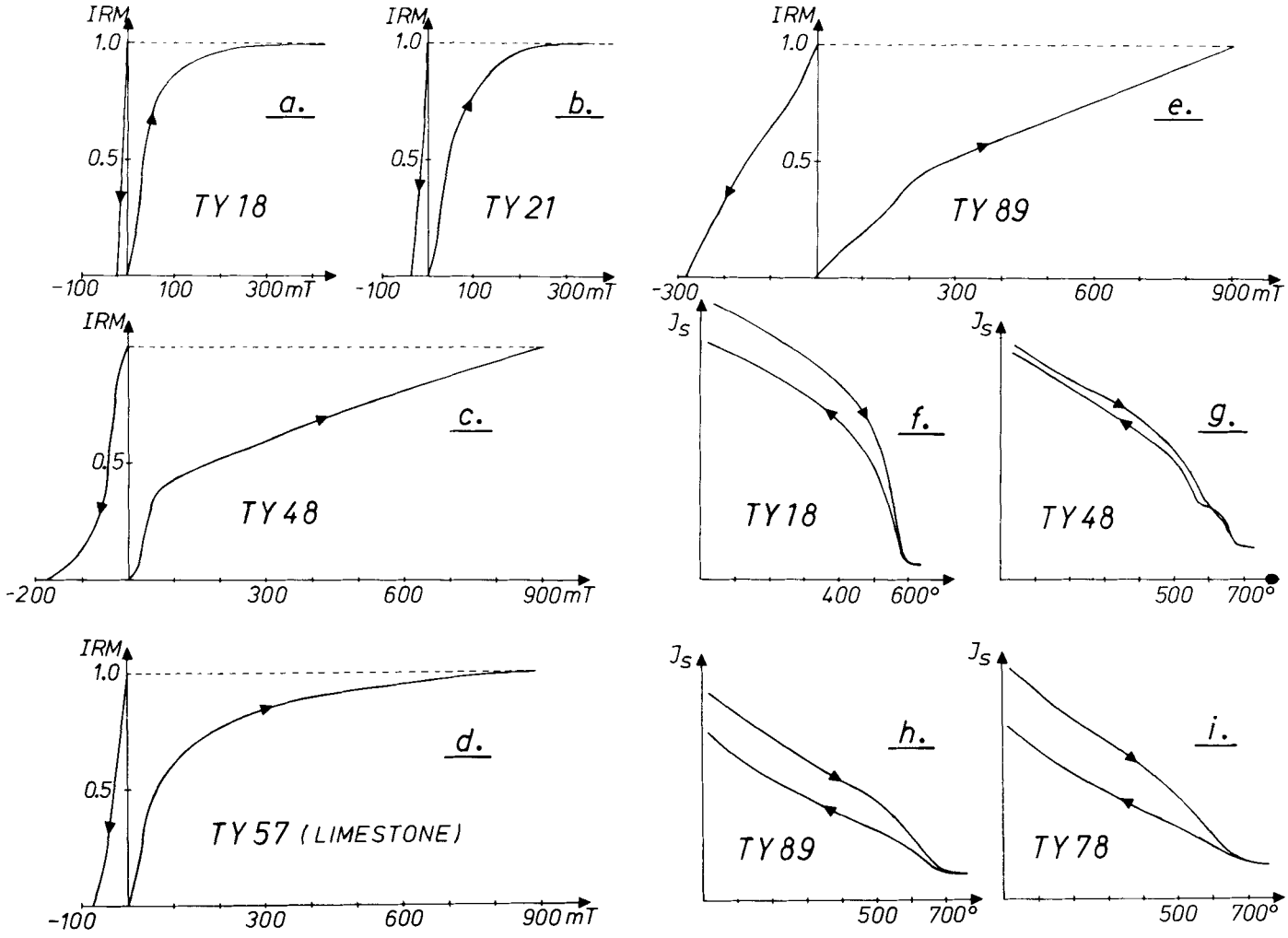


Figure 7. (a–e) Acquisition and back-field isothermal remanence (IRM) curves normalized to the highest applied field, and (f–i) saturation magnetization vs. temperature (J_s–T) curves obtained in a 0.9 T field, for representative samples from Glensaul (TY 18 and 21) and Tourmakeady (all other samples). All except TY 57 are quartz porphyry. For three of the samples (TY 18, 48, 89) both IRM and J_s are shown: the respective curve pairs are indicative of predominantly magnetite (curves a and f), or both magnetite and haematite (curves c and g), or predominantly haematite (curves e and h). Further details in text.

depicts, in somewhat simplified form, the results of analyses of Ordovician palaeomagnetism in Scotland (Carmichael & Storetvedt 1981; Storetvedt & Deutsch 1986). These authors attributed the observed large discordances in the published data set north of the British suture to two main causes: (1) the presence of unresolved polyphase magnetizations in some studies, and/or (2) the recording of appreciable apparent polar wander (APW). The APW history inferred from Storetvedt and Deutsch (1986) involved at least three different palaeomagnetic axes, the final phase being the

Table 4. Middle Ordovician–Silurian palaeomagnetic poles referred to in Fig. 8(a). O_m, O_u and S denote Middle Ordovician, Upper Ordovician and Silurian respectively.

Pole	Formation	Magn. age	Pole	Reference	Remarks
1	Mweelrea ignimbrites, W Ireland	O_m	216 E, 10 N	Morris *et al.* 1973	
2	Remag. of Prec. pluton, Askøy, W Norway	O_{m-u}	233 E, 25 N	Rother *et al.* 1987	Magn. age based on K/Ar evidence
3	Pre-tectonic overprint, Dalradian tillites, Scotl.	O_{m-u}	221 E, 25 N	Stupavsky *et al.* 1982	Pole slightly modified and inferred to be O_{m-u} by Storetvedt & Deutsch, 1986
4	Peterhead granite, Scotl.	S	178 E, 21 N	Torsvik 1985	
5	Foyers granite, Scotl.	S	167 E, 27 N	Torsvik 1984	

Downloaded from https://academic.oup.com/gji/article/92/2/223/752888 by Curtin University Library user on 26 May 2026

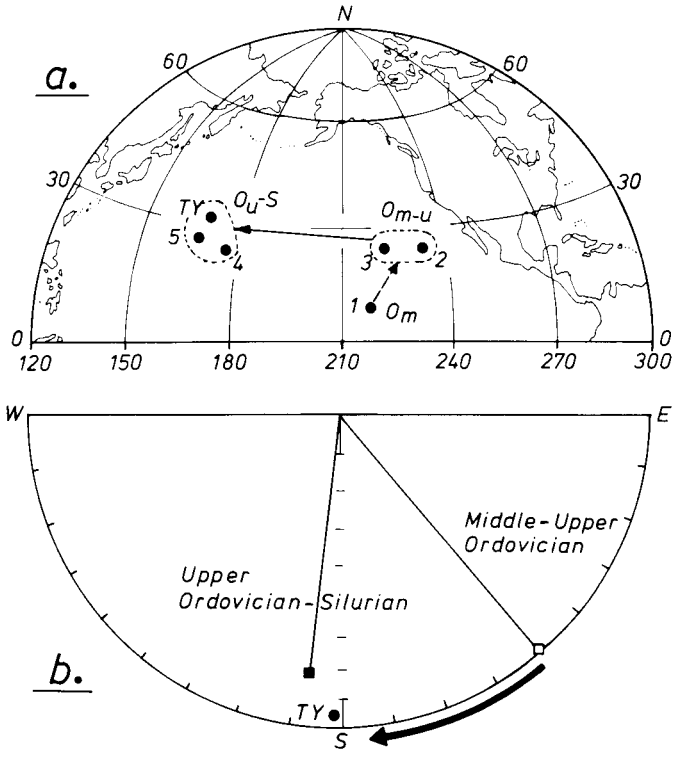


Figure 8. Diagram (a) shows Middle and Upper Ordovician (O_m, O_u) to Silurian (S) north poles reported from areas of the Caledonides north of the presumed collision suture. Age symbols denote magnetization ages cited in original references. Pole numbers refer to those listed in Table 4. TY represents the Tourmakeady volcanic succession, NW Ireland (this paper). Proposed 'O_{m-u}' and 'O_u–S' age groups are shown in dotted outlines which have no statistical significance. Diagram (b) shows for comparison the mid-Ordovician to Silurian mean remanence direction (squares) and corresponding Palaeo-declination axes proposed for Scotland (slightly modified from fig. 4(b) in Storetvedt & Deutsch 1986). The 'Middle to Upper Ordovician' direction (open square), when inverted, corresponds to pole 3 in diagram (a). The 'Upper Ordovician–Silurian' direction (closed square) combines five NW Scottish results discussed in Storetvedt & Deutsch (1986), including numbers 4 and 5 in diagram (a). The Tourmakeady (TY) direction (circle) is shown for comparison. The heavy arrow depicts a *c.* 50° clockwise shift of the SE axis to its SSW position, probably during the late Ordovician; in diagram (a) this shift corresponds closely to the displacement (long arrow) between the 'O_{m-u}' and 'O_u–S' poles. An earlier (mid-Ordovician ?) displacement from pole 1 to poles 2 and 3 may also apply (dashed arrow). Diagram conventions (diagram b) as in Fig. 2.

clockwise axial shift depicted in Fig. 8(b), which probably occurred in the late Ordovician. The scheme of Fig. 8(b) is consistent with the earlier recognition (Deutsch 1980) of two palaeo-declination sets (N–S, NW–SE) in the Ordovician results from both sides of the suture in Britain and Ireland, requiring an anticlockwise rotation of the British Isles relative to the geomagnetic field during the Ordovician. The remanence differences north and south of the suture are mainly in inclination and have been discussed in terms of the width of an Ordovician ocean relative to Ireland (Deutsch 1980, 1984) and Britain (Briden *et al.* 1984). Results from south of the suture in the British Isles are not directly relevant to the present argument and will therefore not be discussed further.

Pole TY, from the present study, is in agreement with Silurian poles 4 and 5 (Fig. 8a). This is consistent with a late Ordovician to Silurian magnetization age for the Tourmakeady–Glensaul rocks (see also below), an age range that falls within the time span of the Taconic orogeny. Pole TY is widely separated from the mid-Ordovician Mweelrea ignimbrite pole (1). If we accept a late Ordovician to Silurian age for our Tourmakeady–Glensaul magnetization, then a large amount of 'late Ordovician' apparent polar wander from pole 1 to pole TY is required. While the actual APW path may have been in two stages (short arrow followed by long arrow in Fig. 8a), the net polar displacement is again about 50° westward as in the case of Scotland. This means that essentially the same clockwise shift between the mid-Ordovician and late Ordovician to Silurian geomagnetic field directions that was demonstrated in Scotland (Fig. 8b) has now been recognized also in Ireland.

Morris' (1976) original study of the Glensaul quartz porphyry yielded two distinct sets of remanence directions, groups 'A' and 'B', which are broadly equivalent to our *A*- and *B*-components, respectively; the comparison is not exact since Morris used AF demagnetization while we have based our results on thermal treatment (cf. discussion above). For group A his *in situ* mean direction (188, +10) is close to our TY-direction (Fig. 8b). However, in his palaeomagnetic and tectonic discussion Morris assumed that his *A*-magnetization is of pre-folding age, an assumption that is not supported by the present study (see also below).

As outlined above, the origin of the investigated rocks has remained a matter of speculation, and geological evidence has apparently been unable to distinguish between an extrusive and an intrusive mode of emplacement. If the extrusive alternative applied, the quartz porphyry would have a Lower Ordovician (Arenig) age. In that case one would expect extensive low-temperature (weathering) alteration of the original magnetite in mid-Ordovician and later times. However, such features have been difficult to find under the microscope, whereas the optical evidence favours a high-temperature origin of the haematite. Also, as follows from Fig. 8, the inferred magnetization age is post-Arenig. We believe, therefore, that the combined palaeomagnetic and rock magnetic evidence points to a late-tectonic (Taconic) intrusive origin of the quartz porphyry. This model implies that all 'felsite' bodies in the Glensaul–Tourmakeady succession are of intrusive origin, and therefore significant 'regional' reheating may have been important. This would lead to slower cooling and somewhat delayed magnetization processes. In view of the low blocking temperatures of the limestones (which are of Arenig age) their observed magnetization was most likely acquired during the post-intrusive cooling of the region.

Despite the large difference in mean strike direction between the two inliers, their *in situ* remanence directions (see Table 3a) are not significantly different. This indicates that the bending of the Glensaul–Tourmakeady structure into an arc pre-dates the intrusive stage.

ACKNOWLEDGMENTS

At Galway, we are indebted to Professor Paul Ryan for crucial geological advice in the field and in later discussion;

Downloaded from https://academic.oup.com/gji/article/92/2/223/752888 by Curtin University Library user on 26 May 2026

to Professor Andrew Brock for logistical support; and to John McKenna who assisted with the rock collection. We are grateful also to: Dr Jagat Prasad who did most of the magnetic measurements; to Per Fluge, Elisabeth Moore, Ray Pätzold and Harald Walderhaug for laboratory assistance; to Tom Vallis for sample preparation; to Knut Breyholtz and Frank Cleveland for the diagrams; and to Anne Berge for typing the manuscript. Constructive comments by Professor N. Bonhommet and an anonymous reviewer are greatfully acknowledged. This work was supported by the Natural Sciences and Engineering Research Council of Canada through Operating Grant A-1946 and Infrastructure Grant A-2677 to E. R. Deutsch.

REFERENCES

Briden, J. C., Morris, W. A. & Piper, J. D. A., 1973. Palaeomagnetic studies in the British Caledonides – VI Regional and global implications, *Geophys. J. R. astr. Soc.*, **34,** 107–134.

Briden, J. C., Turnell, H. B. & Watts, D. R., 1984. British palaeomagnetism, Iapetus Ocean, and the Great Glen fault, *Geology*, **12,** 428–431.

Carmichael, C. M. & Storetvedt, K. M., 1981. Multicomponent magnetization of the Ordovician age younger gabbros, Aberdeenshire, Scotland, *Geophys. J. R. astr. Soc.*, **67,** 199–215.

Deutsch, E. R., 1980. Magnetism of the mid-Ordovician Tramore volcanics, SE Ireland, and the question of a wide Proto-Atlantic Ocean, *J. Geomagn. Geolectr.*, **32,** Suppl. III, 77–98.

Deutsch, E. R., 1984. Mid-Ordovician paleomagnetism and the Proto-Atlantic Ocean in Ireland, in *Plate Reconstruction from Paleozoic Paleomagnetism,* eds Van der Voo, R., *et al., Am. Geophys. Un., Geodyn. Ser.* **12,** 116–119.

Deutsch, E. R. & Somayajulu, C., 1970. Palaeomagnetism of Ordovician ignimbrites from Killary Harbour, Eire, *Earth planet. Sci. Lett.*, **7,** 337–345.

Dewey, J. F. 1961. A note concerning the age of the metamorphism of the Dalradian rocks of western Ireland, *Geol. Mag.*, **98,** 399–405.

Dewey, J. F. 1963. The Lower Palaeozoic stratigraphy of central Murrisk, Co. Mayo, Ireland, and the evolution of the South Mayo trough, *Q. Jl. geol. Soc. Lond.*, **119,** 313–344.

Dewey, J. F., Rickards, R. B. & Skevington, D., 1970. New light on the age of Dalradian deformation and metamorphism in western Ireland, *Norsk geol. Tiddskr.*, **50,** 19–44.

Gardiner, C. I. & Reynolds, S. H., 1909. On the igneous and associated sedimentary rocks of the Tourmakeady district (County Mayo), *Q. Jl. geol. Soc. Lond.*, **65,** 104–153.

Gardiner, C. I. & Reynolds, S. H., 1910. The igneous and associated sedimentary rocks of the Glensaul district (County Galway), *Q. Jl. geol. Soc. Lond.*, **66,** 253–270.

Graham, J. R., Leake, B. E. & Ryan, P. D., 1985. *The Geology of South Mayo* (map, 1:63 360), University of Glasgow.

Lowrie, W. & Heller, F. 1982. Magnetic properties of marine limestones, *Rev. Geophys. Space Phys.*, **20,** 171–192.

Morris, W. A., 1976. Paleomagnetic results from the lower Paleozoic of Ireland, *Can. J. Earth Sci.*, **13,** 294–304.

Morris, W. A., Briden, J. C., Piper, J. D. A. & Sallomy, J. T., 1973. Palaeomagnetic studies in the British Caledonides – V miscellaneous new data, *Geophys. J. R. astr. Soc.*, **34,** 69–105.

Phillips, W. E. A., Stillman, C. J. & Murphy, T., 1976. A Caledonian plate tectonic model, *J. geol. Soc. Lond.*, **132,** 579–609.

Rother, K., Fluge, P. R., Storetvedt, K. M. & Askvik, H., 1987. Palaeomagnetism of the Askøy mafic pluton (late Precambrian), West Norway; events of Caledonian metamorphic remagnetization, *Phys. Earth planet. Int.*, **45,** 85–96.

Ryan, P. D., 1986. Private communication.

Ryan, P. D. & Archer, J. B., 1978. The Lough Nafooey fault: a Taconic structure in western Ireland, *Bull. geol. Surv. Ireland,* **2,** 255–264.

Storetvedt, K. M. & Deutsch, E. R., 1986. Reappraisal of Scottish Ordovician palaeomagnetism, *Geophys. J. R. astr. Soc.*, **87,** 1193–1206.

Stupavsky, M., Symons, D. T. A. & Gravenor, C. P., 1982. Evidence for metamorphic remagnetization of upper Precambrian tillite in the Dalradian supergroup of Scotland, *Trans. R. Soc. Edinb.*, **73,** 59–65.

Torsvik, T., 1984. Palaeomagnetism of the Foyers and Strontian granites, Scotland, *Phys. Earth planet. Int.*, **36,** 163–177.

Torsvik, T., 1985. Palaeomagnetic results from the Peterhead granite, Scotland; implications for regional late Caledonian magnetic overprinting, *Phys. Earth planet. Int.*, **39,** 108–117.

Downloaded from https://academic.oup.com/gji/article/92/2/223/752888 by Curtin University Library user on 26 May 2026